Factors, Multiples, Primes, and Divisibility — Grade 5 Answer Key

Solutions are keyed to the student workbook's page and problem IDs. Equivalent methods are acceptable when the reasoning and result are correct.

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- **P01-01:** Factor pairs: 1×36, 2×18, 3×12, 4×9, 6×6. Checking 1 through 6 finds these pairs; 5 does not divide 36. Any pair after 6×6 only reverses a pair already found.
- **P01-02:** Factor pairs: 1×48, 2×24, 3×16, 4×12, 6×8. There are 10 positive factors: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. Checking 1 through 6 finds these pairs; 5 does not divide 48. Both factors cannot be at least 7, since 7×7 = 49 is greater than 48, so the list is complete.

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- **P02-01:** 1, 2, 3, 4, 6, 8, 12, 24. Division checks: 24÷1 = 24, 24÷2 = 12, 24÷3 = 8, 24÷4 = 6, 24÷6 = 4, 24÷8 = 3, 24÷12 = 2, 24÷24 = 1.
- **P02-02:** 1, 2, 3, 6, 9, 18, 27, 54. Division checks: 54÷1 = 54, 54÷2 = 27, 54÷3 = 18, 54÷6 = 9, 54÷9 = 6, 54÷18 = 3, 54÷27 = 2, 54÷54 = 1.

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- **P03-01:** 9, 18, 27, 36, 45, 54, 63, 72.
- **P03-02:** Yes. 7 × 12 = 84.

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- **P04-01:** Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. The common factors are **1, 2, 3, and 6**.
- **P04-02:** Factors of 28: 1, 2, 4, 7, 14, 28. Factors of 42: 1, 2, 3, 6, 7, 14, 21, 42. The common factors are **1, 2, 7, and 14**.

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- **P05-01:** GCF(24, 36) = **12**. Method 1: the common factors are 1, 2, 3, 4, 6, 12, and 12 is the greatest. Method 2: 24 = 2×2×2×3 and 36 = 2×2×3×3. The shared prime factors, counting repeats, are 2, 2, and 3, whose product is 12.
- **P05-02:** GCF(45, 60) = 15 kits. Each kit has 45 ÷ 15 = 3 pencils and 60 ÷ 15 = 4 erasers.

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- **P06-01:** 1,240 is divisible by 2, 5, and 10. 1,275 is divisible by 5 only. 1,286 is divisible by 2 only.
- **P06-02:** 1,000. It must end in 0 to be divisible by both 2 and 5; 1,000 is the least four-digit number ending in 0.

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- **P07-01:** Digit sum 4+5+7+2 = 18. It is divisible by both 3 and 9.
- **P07-02:** For 63[ ], the digit sum is 9 + [ ]. It is divisible by 9 when that sum is 9 or 18, so the possible digits are **0 or 9**.

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- **P08-01:** Yes. The last two digits are 16, and 16 ÷ 4 = 4.
- **P08-02:** Yes. The last three digits are 248, and 248 ÷ 8 = 31.

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- **P09-01:** 1 is neither; 17 and 29 are prime; 21 and 35 are composite.
- **P09-02: 41, 43, 47, 53, and 59**. Rule out even numbers and multiples of 5. The other composites are 49 = 7×7, 51 = 3×17, and 57 = 3×19. The five listed primes are not divisible by 2, 3, 5, or 7. These checks suffice: both factors of a composite below 60 cannot be at least 8, because 8×8 = 64, and divisibility by 4 or 6 would already be caught by the test for 2.

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- **P10-01:** 72 = **2×2×2×3×3** = **2^3×3^2**. Check: 8×9 = 72.
- **P10-02:** 210 = **2×3×5×7**. Check: 2×3 = 6, 6×5 = 30, and 30×7 = 210.

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- **P11-01:** One tree splits 84 into 2 and 42, then 42 into 2 and 21, then 21 into 3 and 7. Its prime leaves give 84 = **2×2×3×7** = **2^2×3×7**. Other correct trees are acceptable.
- **P11-02:** One tree splits 90 into 9 and 10, then 9 into 3 and 3 and 10 into 2 and 5. Another splits 90 into 2 and 45, then 45 into 5 and 9, then 9 into 3 and 3. Both have the prime leaves 2, 3, 3, and 5, giving **2×3×3×5** = **2×3^2×5**. Different splits only regroup the factors; every correct tree ends with the same prime factors, possibly in a different order.

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- **P12-01:** Multiples of 6: 6, 12, 18, 24; of 8: 8, 16, 24. LCM = **24**.
- **P12-02:** 12 = 2^2×3 and 15 = 3×5, so LCM = 2^2×3×5 = **60**.

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- **P13-01:** LCM(4, 6) = 12, so they flash together after **12 seconds**.
- **P13-02:** LCM(10, 15) = 30 minutes; the next arrival is **8:30 a.m.**

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- **P14-01:** Possible arrays include 1×24, 2×12, 3×8, and 4×6 (and their turned versions).
- **P14-02:** Side-length pairs are **1×48, 2×24, 3×16, 4×12, and 6×8**.

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- **P15-01:** [ ] = 13, because 117 ÷ 9 = 13; check 9×13 = 117.
- **P15-02:** The number is **32**, because a positive whole number is its own greatest positive factor. Its factor pairs are 1×32, 2×16, and 4×8. Testing 1 through 5 finds these pairs; 3 and 5 do not divide 32. Both factors cannot be at least 6, since 6×6 = 36 is greater than 32, so there are no missing factors.

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- **P16-01: 105 and 120**. Numbers divisible by both 3 and 5 are multiples of 15. The nearby multiples are 90, 105, 120, and 135, so only 105 and 120 lie from 100 to 130. Checks: 105÷3 = 35, 105÷5 = 21; 120÷3 = 40, 120÷5 = 24.
- **P16-02: 36 and 72**. The two-digit numbers with digit sum 9 are 18, 27, 36, 45, 54, 63, 72, 81, and 90. Only 36 = 4×9 and 72 = 4×18 are divisible by 4. Both are composite, and 3+6 = 7+2 = 9.

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- **P17-01:** GCF(32, 48) = 16. There are **16 teams**, with 2 students and 3 parents per team.
- **P17-02:** GCF(72, 90) = 18. There are **18 boxes**, with 4 chocolate and 5 vanilla cookies per box.

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- **P18-01:** 157 ÷ 12 = **13 remainder 1**; $12 \times 13 + 1 = 157$.
- **P18-02:** 98 ÷ 8 = 12 remainder 2, so **13 vans** are needed; the two remaining children require another van.

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- **P19-01:** Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Their greatest shared factor is **6**, the GCF. Multiples of 18: 18, 36, 54, 72; multiples of 24: 24, 48, 72. Their smallest shared positive multiple is **72**, the LCM.
- **P19-02:** Divisible by 2 (even), 3 (digit sum 9), 4 (last two digits 08), 8 (last three digits 008), and 9 (digit sum 9). Not divisible by 5 or 10 because it does not end in 0 or 5.

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- **P20-01:** **$30 = 2 \times 3 \times 5$**. The three smallest distinct primes are 2, 3, and 5. Any number with three distinct prime factors must contain at least these three smallest primes or larger ones; adding repeated factors or using larger primes increases the product. Thus 30 is the smallest. Its factor pairs are 1×30, 2×15, 3×10, and 5×6, so all its positive factors are **1, 2, 3, 5, 6, 10, 15, and 30**.
- **P20-02:** False: 2 is even and prime. Correct statement: **Every even number greater than 2 is composite**, because it has factors 1, 2, and itself.